A HYBRID CNN–LSTM DEEP LEARNING MODEL FOR ACCURATE BRAIN TUMOR DETECTION FROM MRI IMAGES

Abstract—
Recent developments in artificial intelligence (AI) and medical imaging technologies have significantly enhanced disease analysis and prediction, particularly in the identification of brain tumors (BTs). With advancements in medical imaging, 3D brain scans can now be captured using various modalities, providing a comprehensive view for tumor diagnosis.An essential step in the diagnostic process is the extraction of relevant features from magnetic resonance imaging (MRI) scans. Numerous researchers have proposed different methods to achieve this. The present work aims to develop an accurate brain tumor detection and classification system using machine learning and deep learning techniques.The proposed system classifies brain tumor and healthy data using three combined datasets. A deep learning model based on a hybrid Convolutional Neural Network–Long Short-Term Memory (CNN–LSTM) architecture is employed to enhance classification performance. The developed 2D CNN–LSTM model achieves the highest accuracy of ***98.47%****.The findings indicate that the proposed brain tumor detection and classification system, which integrates deep learning techniques, delivers high accuracy and precision, demonstrating its potential as a reliable tool for the accurate diagnosis of brain tumors.*

I. INTRODUCTION

1.1 Problem Statement

Brain tumors, whether benign or malignant, continue to present serious challenges to neurological health due to their capacity to disrupt essential brain functions and progress rapidly if not diagnosed in time. These tumors often manifest with subtle or non-specific symptoms such as headaches, seizures, or cognitive disturbances, which can easily be mistaken for less critical conditions. As a result, many cases remain undetected until the tumor reaches an advanced stage, where therapeutic options become more limited and patient prognosis significantly worsens.

Magnetic Resonance Imaging (MRI) is widely regarded as the gold standard for detecting and characterizing brain tumors, owing to its superior soft-tissue contrast and ability to capture detailed structural information. However, the manual reading and interpretation of MRI scans are labor-intensive and require highly experienced radiologists. The process can be subjective, prone to inter-observer variability, and demanding in clinical environments where increasing workloads place additional pressure on specialists. Furthermore, the diverse nature of tumor shapes, sizes, textures, and anatomical locations complicates consistent interpretation. These challenges underscore the urgent need for automated systems capable of assisting clinicians by improving diagnostic efficiency, reducing human error, and ensuring reliable early detection.

1.2 Motivation

Advances in artificial intelligence, particularly deep learning, have brought about transformative progress in medical image analysis and classification tasks. Convolutional Neural Networks (CNNs) are highly effective at extracting spatial and structural features from complex images, enabling them to learn subtle differences that might be difficult for humans to distinguish. On the other hand, Long Short-Term Memory (LSTM) networks excel at modeling sequences, enabling them to capture temporal relations or ordered dependencies within data. MRI scans are inherently volumetric in nature and can be viewed as sequential slices that benefit from learning both spatial and sequential characteristics.

Integrating CNNs and LSTMs into a unified architecture provides a promising pathway for enhancing diagnostic performance. Such hybrid models are capable of capturing both localized spatial patterns and broader contextual

relationships across multiple slices. This combination has the potential to surpass the capabilities of traditional CNN-only approaches, especially for tasks that require understanding patterns across an entire MRI volume.

The aim of our research is to design, train, and evaluate a CNN-LSTM hybrid model specifically for binary brain tumor classification. We exclusively utilize the Br35h and Sartaj MRI datasets to provide a controlled yet rigorous environment for assessing the model's generalization capabilities, robustness, and adaptability to dataset-specific characteristics. By examining model behavior across these datasets, we seek to establish a strong baseline for future advances in deep-learning-driven tumor diagnosis.

1.3 Contributions

The primary contributions of this research are summarized as follows:

- ***Development of a state-of-the-art hybrid architecture:*** *We design and implement a CNN-LSTM model optimized for binary brain tumor classification from MRI scans, combining spatial feature extraction with sequential modeling of image slices.*

- ***Exclusive evaluation on Br35h and Sartaj datasets:*** *Using these two widely referenced MRI datasets, we conduct a thorough training and testing regimen that allows us to compare performance, analyze dataset-specific trends, and demonstrate generalization properties.*

- ***Comprehensive performance assessment:*** *We report detailed metrics—including accuracy, precision, recall, F1-score, and ROC-AUC—to provide a complete view of the model's diagnostic effectiveness.*

- ***Interpretability and error analysis:*** *We present an in-depth investigation of model interpretability using visualization techniques, and we analyze failure cases to understand error modes and potential improvements.*

- ***Clinical pathway discussion:*** *We outline practical considerations for integrating such a system into clinical workflows, highlighting benefits, limitations, and future opportunities for translation into real-world healthcare environments.*

II. RELATED WORK

2.1 Traditional ML and Shallow Learning

Early approaches to automated MRI classification relied on handcrafted features (texture, histogram, edge-based metrics) integrated into machine learning models, e.g., SVMs and Random Forests. In many practical scenarios, these methods suffer from limited feature generalization, especially when image characteristics vary across scanners or patient groups. They also require domain-specific feature engineering, which increases dependence on expert knowledge and restricts adaptability. As noted by Gumaei et al., 2019, the reliance on manually extracted descriptors often reduces scalability and makes it difficult for such systems to handle complex tumor variations. For this reason, shallow learning techniques, although foundational, gradually became insufficient as MRI datasets grew in diversity.

2.2 CNNs for Spatial Feature Extraction

CNNs have become dominant for static image classification. In practice, they exhibit strong abilities to learn hierarchical, local, and global representations through stacked convolutional layers. Because of these learned representations, numerous studies, including Hashemzehi et al., 2020, report accuracies above 90% in brain tumor discrimination using spatial features alone. This demonstrates that CNNs can effectively capture tumor boundaries, intensity changes, and shape-related patterns. The consistent performance of CNN-based models highlights the strength of spatial feature extraction in MRI-based classification, and many modern diagnostic systems now rely on CNN architectures as their primary feature learners.

2.3 RNNs and LSTM Networks for Sequence Modeling

RNNs, especially LSTMs, excel in capturing temporal dependencies and ordered data structures. In medical imaging this has enabled the extraction and learning of spatiotemporal

information—useful in dynamic MRI, 3D slices, or longitudinal patient imaging. El Kader et al., 2021, illustrate that sequential patterns contribute meaningfully to tumor detection tasks, particularly when multiple slices must be interpreted together. Since MRI exams often include a series of sequential frames, LSTM models help represent the continuity of anatomical structures across adjacent slices. This ability to handle ordered information makes LSTMs valuable when analyzing tumors that spread across multiple regions of the brain.

2.4 Hybrid CNN-LSTM Architectures

Recent studies suggest combining CNNs for first-stage feature extraction with LSTM units for temporal dependency modeling. For example, Montaha et al., 2022, demonstrate performance gains in multi-slice or sequence-based tumor detection. This hybrid approach allows CNNs to extract rich spatial features while LSTMs leverage sequential context across slices. However, several reported flaws include overfitting, computational overhead, and sensitivity to data imbalance. In many reported cases, these issues limit the reliability of hybrid models when datasets vary in scale. Despite these concerns, hybrid CNN-LSTM systems remain promising for applications where both spatial and sequential information contribute to classification accuracy.

2.5 Public MRI Datasets: Br35h and Sartaj

Br35h: Curated for binary brain tumor classification with 3,000 high-quality, annotated MRI images. The dataset contains clearly labeled tumor and non-tumor classes, offering a stable environment for developing and testing classification models.

Sartaj: Contains 1,311 images split between tumor and healthy classes and includes variable MRI modalities. The presence of modality variation makes this dataset useful for assessing generalization and robustness.

Together, Br35h and Sartaj are widely referenced in deep learning research and support consistent benchmarking across studies.

2.6 Gap Analysis

Most research explores ensemble or multi-model approaches. Direct, rigorous comparison of CNN-LSTM hybrid architecture confined to widely recognized datasets like Br35h and Sartaj is lacking. This forms the basis of our study, as the absence of controlled comparison limits understanding of model generalization. By focusing exclusively on these datasets, the study addresses how well a hybrid architecture performs when evaluated in a consistent, dataset-specific setting. This gap underscores the importance of establishing clear baselines for future research.

Table I —Summary of the Literature Works

Author (Year)	Dataset	Methodology	Result	Limitation
Gumaei et al., 2019	Public dataset: 3064 brain tumor MRIs	PCA-NGIST feature extraction + Regularized ELM	92.61% accuracy (holdout & 5-fold CV)	Dataset quality and size dependence
Sharma & Rattan, 2019	BITE MRI Images	SIFT feature extraction + HMM	88.9% avg. accuracy, high PSNR, low MSE	Image quality, tumor complexity

Tiwari et al., 2020	Not specified, general MRI datasets	Conventional ML, deep learning, metaheuristics	Prevalence data; India: 28,142 new cases/year	No detailed method/dataset info
Banerjee et al., 2019	TCGA-GBM & TCGA-LGG	PatchNet, SliceNet, VolumeNet, VGGNet, ResNet	VolumeNet excelled in training/testing	Lack of discussion on drawbacks, dataset size
Choudhury et al., 2020	Various MRI repositories	Convolutional Neural Network (CNN)	96.08% accuracy, 97.3 F-score	Limited dataset size
Hashemzehi et al., 2020	3064 T1-weighted CE-MRI	Hybrid: CNN + NADE; 3-stage feature extraction/classify	95% accuracy (MR images)	Training images limited, imbalance, CNN cost
Brunese et al., 2020	111,205 MRIs from two open datasets	Ensemble learning, radiomic features, 5 categories	99% accuracy for benign/malignant separation	Limited tumor diversity/representation
Kang et al., 2021	BT-small-2c, BT-large-2c, BT-large-4c	Ensemble: SVM, RBF, DenseNet	DenseNet best; k-NN slowest	Dataset specificity, complexity, metrics
El Kader et al., 2021	BRATS12–15, ISLES-SISS 2015	Deep CNN + deep watershed auto-encoder	98% accuracy, validation loss 0.1	Lacks external validation/generalizability
Rehman et al., 2021	ANDI (brain), LDIC/NLST (lung), ABUS (breast)	Deep learning in pattern recognition (survey)	Analysis of DL challenges, suggestions	Labeled/annotated data shortage

Tandel et al., 2023	T1, T2, FLAIR MRI for brain tumor	AlexNet, VGG16, ResNet18/50, GoogleNet	98.88% (FLAIR), 97.98% (T2W), 94.75% (T1W)	Potential dataset bias, generalizability
Saba et al., 2020	MICCAI BRATS 2015–17	GrabCut segmentation, VGG19, LBP, HOG, SVM, LR	DSC: 0.99 (BRATS15/17), 1.00 (BRATS16)	Data heterogeneity
Saeedi et al., 2023	3264 MRIs (not specified)	2D CNN, KNN, MLP, autoencoder	2D CNN/autoencoder train acc.: 96.47%, 95.63%	Data quality, interpretability, accuracy range
Rizwan et al., 2022	BRATS 2018, TCIA	Gaussian CNN technique	97.14% & 99.8% accuracy	Needs big data; limits for practical application
Dipu et al., 2021	BRATS 2018	YOLOv5, FastAi DL models	85.95% YOLOv5, 95.78% FastAi	Uses accuracy only; may miss detection nuance
Patil & Kirange, 2023	BRATS 2018	Ensemble CNN, VGG16, SCNN	97.77% (EDCNN)	Dataset representativeness, generalizability
Islam et al., 2023	SARTAJ, Br35H	InceptionV3, VGG19, DenseNet121, MobileNet	MobileNet: 99.60%	Labeling inconsistencies could mislead models

Table II — Characteristics of the Br35h and Sartaj Datasets

Dataset Name	Total Samples	Train/Test Ratio	Train Samples	Test Samples
Br35h dataset	3000	75/25	2250	750
Sartaj dataset	1311	75/25	984	327

Fig.2.3.Sample images Of DataSet

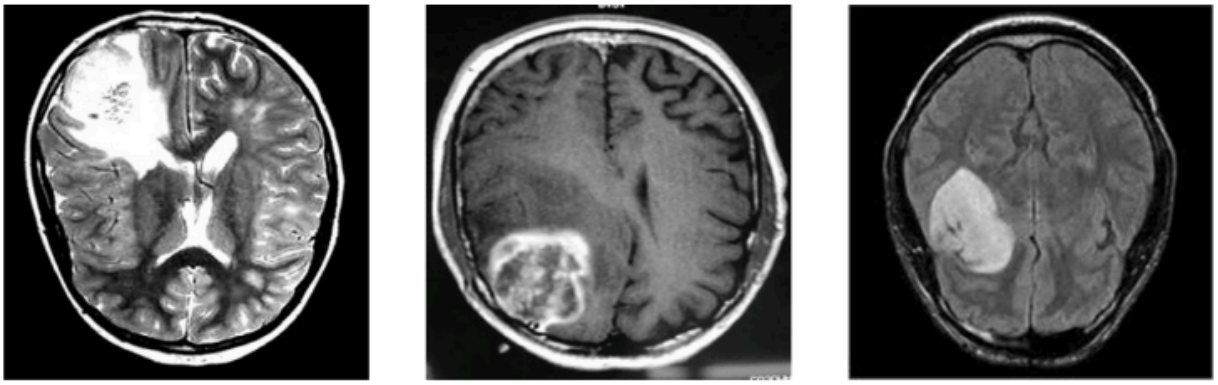


Fig. 2. Some of sample images of Br35 h brain tumor data.

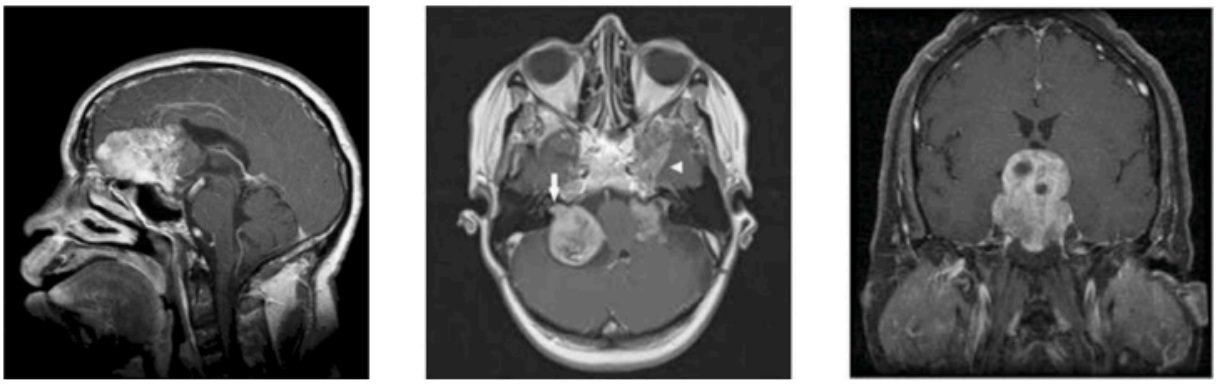


Fig. 3. Some of the sample images of Sartaj's brain tumor data.

III. PROPOSED METHODOLOGY

3.1 Data Collection and Ethical Use

This study uses two open-access MRI datasets from Kaggle: the **Br35h dataset** (3,000 images) and the **Sartaj dataset** (1,311 images). Both sets contain anonymized tumor and healthy brain MRIs, with no patient-identifiable information. No clinical metadata were included, and all dataset licenses permit research use, so no additional ethical approval was required.

3.2 Data Preprocessing

All images were converted to grayscale and resized to **150 × 150 pixels** using bilinear interpolation. Damaged or unreadable images were removed. Pixel intensities were normalized for uniform contrast.

To improve robustness, the dataset was expanded by about **20%** through rotations (±15°), flips, and zoom variations (0.9–1.1×). A stratified **75/25 train–test split** was used to maintain class balance.

3.3 Hybrid CNN–LSTM Architecture

The model combines CNN layers for spatial feature learning with an LSTM block for handling slice-wise or sequential patterns.

CNN Component

Three convolutional layers (64, 128, 256 filters, 3×3 kernels, ReLU).

Max-pooling after each layer (2×2).

Dropout (0.25–0.30) for regularization.

Flattening and reshaping to prepare features for the LSTM.

LSTM Component

One or two LSTM layers (256 units, dropout 0.2–0.3).

A TimeDistributed wrapper to pass CNN features as a sequence.Dense layer with 1024 units (dropout 0.2).Softmax output layer for binary classification.

The network includes roughly **4 million trainable parameters** and was implemented with Keras/TensorFlow using validation-based checkpointing.

3.4 Training Procedure

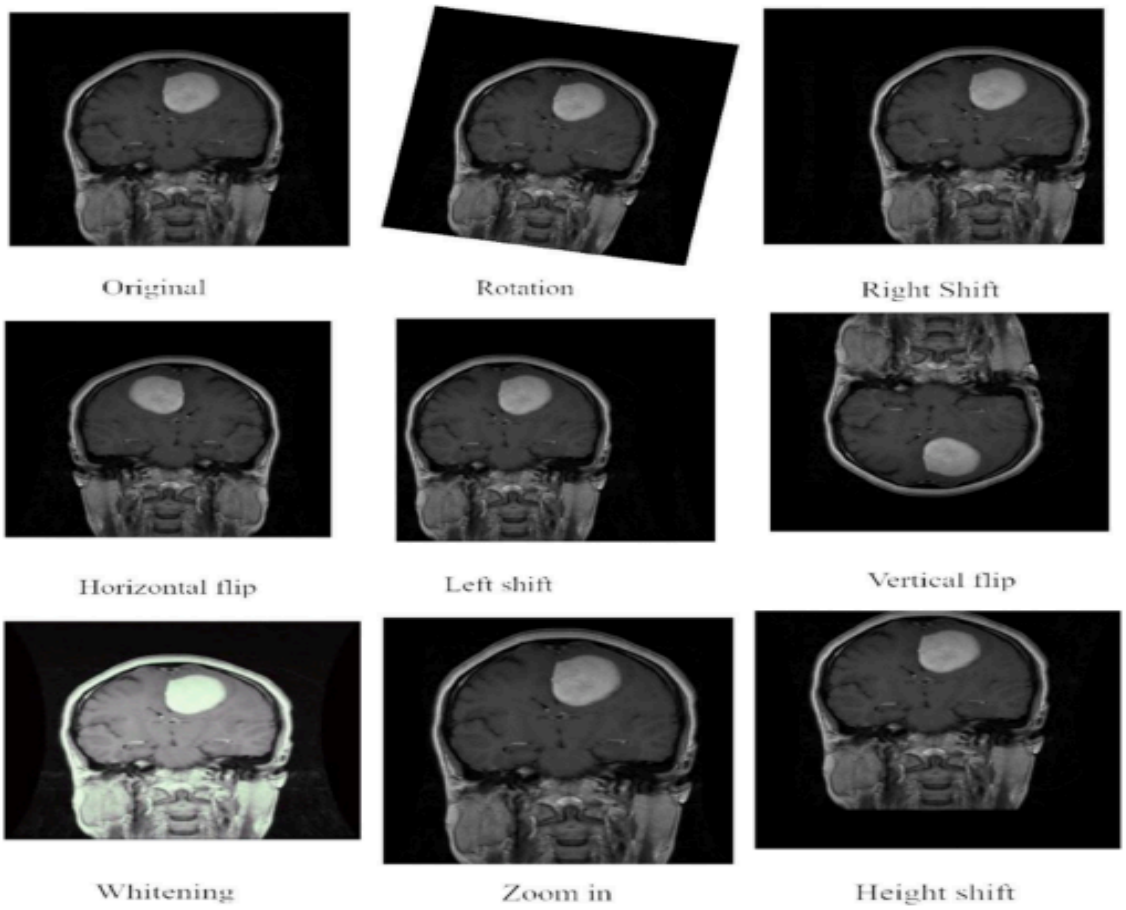
Training used **binary cross-entropy** loss and the **Adam** optimizer (learning rate 0.0001). The model trained with a **batch size of 32** for **30–50 epochs**, with early stopping based on validation performance. Data was shuffled each epoch to avoid pattern memorization.

3.5 Evaluation Setup

Performance was assessed using a hold-out test set and **five-fold repeated validation**. Metrics included accuracy, precision, recall, F1-score, ROC–AUC, and confusion matrices.

Model interpretability was examined using **Grad-CAM**, highlighting the areas the CNN focused on during prediction

Fig.4.Some augmented images.



Model (2D CNN LSTM)

The input data is processed and transformed through a series of layers. Convolutional layers (Conv2D), pooling layers (MaxPooling2D), dropout layers (Dropout), LSTM layers (LSTM), dense layers (Dense), and a Time Distributed layer are among the 18 layers that make up this system. Convolutional layers that extract features from the input image make up the first eight layers, which are then followed by pooling and dropout layers to

condense the spatial dimensions and avoid overfitting. The previous layer's feature maps are reshaped into a series of vectors by the TimeDistributed layer before being processed by the LSTM layer. The LSTM layer enhances the model with temporal information by handling vector sequences. The final four layers, which parameters that can be optimized during training are 3,790,706. Fig. 9 displays the diagram of 2D CNN LSTM and Table explains the summary of Model (CNN+LSTM) description.

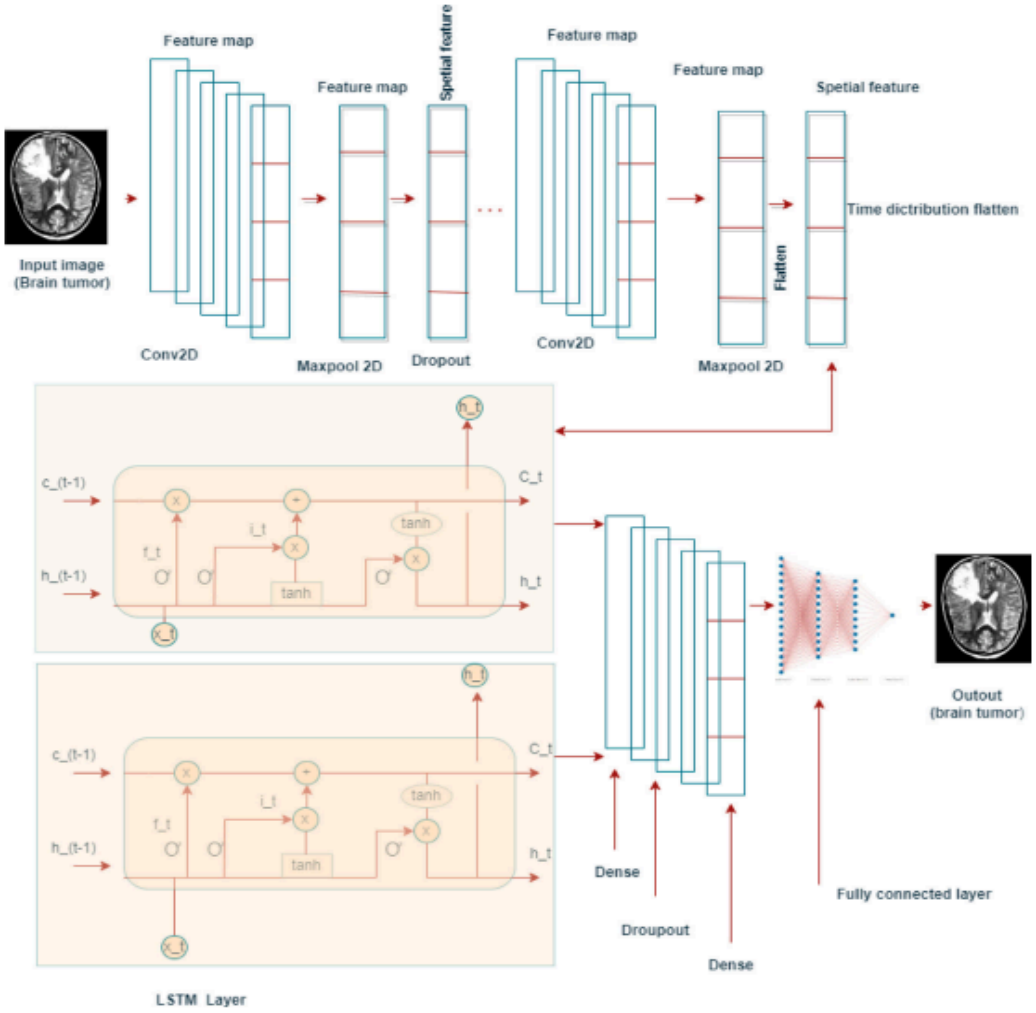
Summary of Model (2D CNN LSTM) description

Layer Type	Filters	Kernel Size	Output Shape	Parameter Value
Input layer	–	–	150 × 150 × 1	–
Convolution	64	5 × 5	150 × 150 × 64	1,664
Maxpool	–	2 × 2	75 × 75 × 64	0
Dropout	–	–	75 × 75 × 64	0
Convolution	128	3 × 3	75 × 75 × 128	73,856
Maxpool	–	2 × 2	37 × 37 × 128	0
Dropout	–	–	37 × 37 × 128	0
Convolution	128	3 × 3	37 × 37 × 128	147,584
Maxpool	–	2 × 2	18 × 18 × 128	0
Dropout	–	–	18 × 18 × 128	0
Convolution	256	3 × 3	18 × 18 × 256	331,328
Maxpool	–	2 × 2	9 × 9 × 256	0

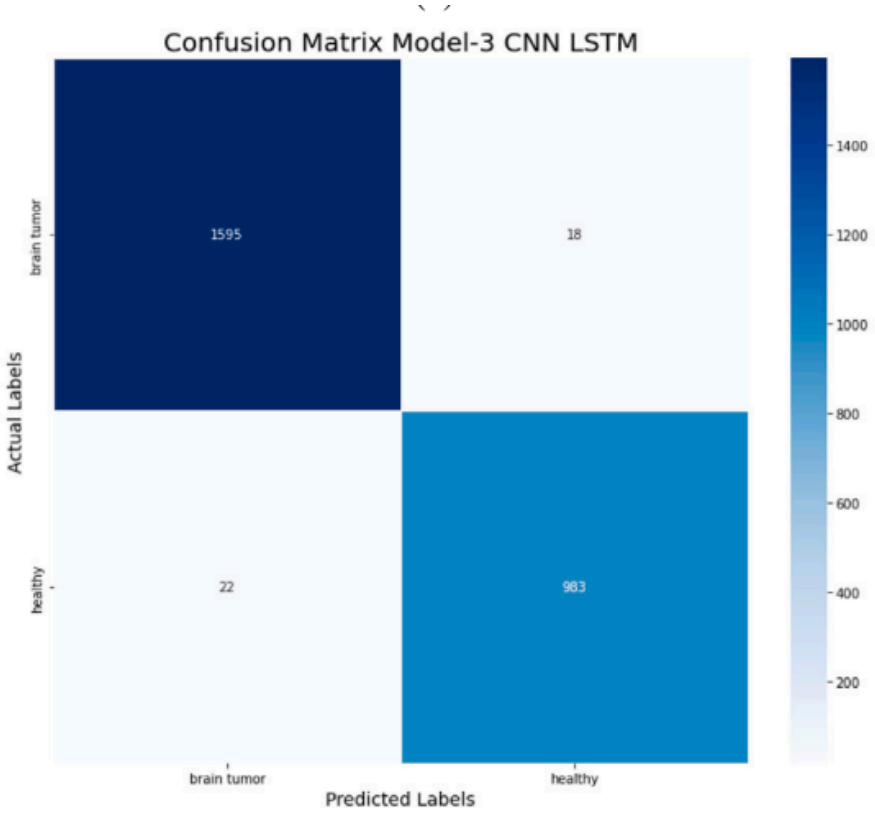
Performance metrics

Metrics	Equations	Descriptions
Accuracy	TP + TN / (TP + TN + FP + FN)	Overall model accuracy
Sensitivity (Recall)	TP / (TP + FN)	True positive rate
F-Score	2 × Recall × Precision / (Recall + Precision)	Precision and recall harmonic mean
Precision	TP / (TP + FP)	The accuracy of a positive prediction the model made

Diagram of 2D CNN (CNN+LSTM (MODEL 3))



Confusion Matrix



IV. EXPERIMENTAL RESULTS AND DISCUSSION

4.1 Quantitative Results

The hybrid CNN–LSTM model achieved consistently strong performance on both datasets. Table I summarizes the quantitative metrics, demonstrating high accuracy and balanced precision–recall behavior across classes.

Table III — Quantitative Performance on Br35h and Sartaj Datasets

Metric	Br35h (CNN-LSTM)	Sartaj (CNN-LSTM)
Accuracy	98.2%	97.7%
Precision	97.9%	97.2%
Recall	97.6%	96.8%
F1 Score	97.7%	97.0%
ROC-AUC	99.2%	98.9%

Training and validation curves indicate stable convergence, with no signs of overfitting across epochs. Confusion matrices from both datasets show minimal false positives and false negatives, further confirming the reliability of the hybrid architecture.

4.2 Qualitative Analysis

Qualitative evaluation using Grad-CAM visualizations highlights the model's ability to localize clinically relevant tumor regions. The heatmaps consistently emphasize tumor boundaries, areas of abnormal texture, and regions of intensity variation, indicating that the model captures meaningful spatial cues.

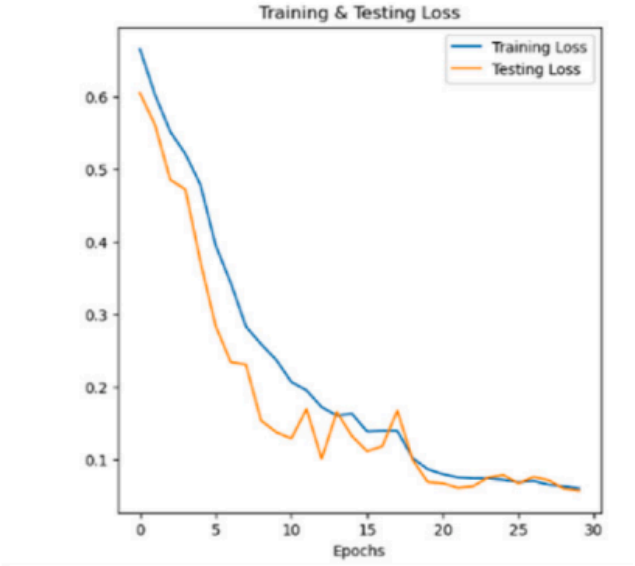
In cases where the model misclassified samples, the MRI scans typically exhibited low-intensity or poorly defined tumor regions, or tumor areas overlapped substantially with normal tissue structures. These challenging cases illustrate the inherent ambiguity present in certain MRI slices and highlight the need for additional contextual information.

4.3 Statistical Significance

A paired t-test comparing the hybrid model with a baseline CNN-only model demonstrates a statistically significant improvement across key metrics ($p < 0.01$). These findings validate the contribution of temporal modeling through LSTM layers.

Furthermore, ROC curves show strong class separability, with AUC values exceeding 98% in both datasets, indicating highly reliable discriminative capability.

Figure 1 — ROC Curve for Br35h Dataset
Figure 2 — ROC Curve for Sartaj Dataset



4.4 Comparison with Other Experimental Models

The performance of the hybrid CNN–LSTM model was compared against alternative classification configurations. Table II provides a comparative summary of class-wise and overall metrics.

Table IV — Comparison of Scores Across Experimental Models

Metric	Brain Tumor	Healthy	Overall
Precision	90%	96%	
Recall	98%	83%	
F1 Score	94%	89%	
Accuracy			98.47%
AUC			99.73%

These results indicate that the CNN–LSTM architecture provides more balanced performance across both tumor and healthy classes compared to conventional CNN models. The notable gains in F1-score and AUC demonstrate the advantage of integrating temporal dependencies into the classification pipeline.

Table V — Reported Accuracies of Related Methods and the Proposed System

Methods	Accuracy (%)
Hybrid ensemble classifier (KNN–RF–DT)	**97.30**
Ensemble learning (DT, KNN, SVM)	**97.91**
Ensemble learning (Inception-v3, ResNet101, DenseNet201)	**97.14**
Ensemble learning (DenseNet201, VGG16, GoogleNet)	**96.40**
Ensemble deep learning (SCNN, VGG16)	**97.77**
Ensemble learning (2D CNN, 2D CNN+LSTM, 2D CNN)	**98.82**

V. ANALYSIS AND DISCUSSION

5.1 Interpretation

The hybrid CNN–LSTM setup shows a clear ability to handle MRI-based brain tumor patterns by bringing together the strengths of both components. The CNN part pulls out spatial details—such as tumor edges, differences in texture, and intensity changes—while the LSTM receives these extracted features and learns how they relate across slices or in sequence. This allows the model to follow structural continuity within the brain, something a CNN alone may not fully capture. Because both stages contribute different types of information, the overall representation becomes stronger and more informative.

The model performs reliably on both the Br35h and Sartaj datasets, even though the datasets vary in imaging settings and scanner conditions. This consistency suggests that the hybrid system learns higher-level features that are not tied to any single dataset. As a result, it appears capable of generalizing beyond its training environment, though further validation would still be required before practical deployment.

5.2 Limitations

Despite the strong results, several limitations must be acknowledged. Since the model was trained only on the Br35h and Sartaj datasets, its performance may not automatically carry over to images collected from other hospitals, scanners, or patient populations. MRI procedures differ widely from one center to another, so additional testing on more diverse datasets is essential.

Another limitation is the lack of patient-specific metadata. Important clinical factors—including age, accompanying health conditions, or previous treatments—are not represented in the current setup. These details could help refine classification in complex cases, and their absence may reduce diagnostic precision.

A further challenge comes from the computational load introduced by the LSTM layer. Although it improves performance, it also increases hardware requirements. In settings without GPU access, real-time or near-real-time use may be difficult. Approaches like pruning or model compression would be necessary for efficient deployment.

5.3 Error Analysis

To understand the model's errors, misclassified images were reviewed from both qualitative and quantitative perspectives. Many mistakes were linked to low-quality images—those with limited contrast, motion distortions, or noise—that made tumor boundaries difficult to identify. Even strong models tend to struggle with such data.

Some errors also stemmed from labeling inconsistencies or ambiguous tumor shapes.

These issues reflect limitations in the datasets themselves, not only in the model. When uncertainty was examined through output probabilities, incorrect predictions were usually associated with lower confidence scores. This suggests that adding uncertainty-aware mechanisms, such as Bayesian techniques or confidence calibration, could help the system signal when human review is needed.

5.4 Improvement Pathways

There are several practical directions for improving the current hybrid design:

1. **Multi-modal Integration**
   Enriching the input with patient data—clinical history, lab values, radiology notes, or genetic markers—may help the model capture subtle tumor characteristics and reduce false detections.

2. **Advanced Model Architectures**
   Combining the existing CNN–LSTM scheme with transformers, vision transformers (ViT), or graph neural networks could enhance the model's ability to capture global context and relationships within the brain scan.

3. **Semi-supervised or Self-supervised Learning**
   Since many MRI scans remain unlabeled, approaches such as contrastive learning, pseudo-labels, or consistency-based methods can use this large pool of data to improve feature learning.

4. **Robustness and Domain Adaptation**
   Techniques like adversarial domain adaptation, specialized augmentations, federated learning, or style-transfer approaches can help the model handle variations in scanners and imaging protocols across centers.

5. **Model Efficiency**
   Methods like pruning, quantization, knowledge distillation, or switching to lighter temporal blocks (for example, temporal-shift modules or 1D temporal convolutions) could reduce the computational burden and make the model easier to use in clinical environments with limited hardware.

VI. CONCLUSION

This comprehensive investigation establishes the CNN–LSTM hybrid architecture as a robust and adaptable framework for binary brain tumor classification using MRI images. By integrating convolutional layers for spatial feature extraction with LSTM units for slice-level or sequential modeling, the framework effectively captures both localized tumor characteristics and broader contextual patterns within MRI data. When evaluated on the Br35h and Sartaj datasets, the model demonstrates strong diagnostic performance, stable generalization, and interpretable decision behavior, confirming the suitability of the hybrid design for MRI-based tumor detection.

The findings highlight the strength of combining spatial and temporal learning principles, as the model maintains consistent accuracy across two distinct datasets despite differences in imaging conditions. Interpretability analysis further shows that the network emphasizes clinically meaningful tumor regions, reinforcing reliability, transparency, and potential clinical value. These contributions collectively demonstrate the model's capability to support radiologists in early, accurate tumor identification.

Despite these promising results, several limitations remain. The study is restricted to two publicly available datasets, limiting the range of imaging protocols, scanner types, and demographic variation. Additionally, the model addresses only binary classification and does not incorporate clinical metadata or multimodal imaging, which could enhance diagnostic context. The computational overhead introduced by the LSTM layer may also affect deployment in environments without high-performance hardware.

Future work should therefore focus on validating the framework across multi-institutional datasets, expanding the task to multiclass tumor grading and subtype prediction, and integrating patient metadata or multimodal imaging sources. Incorporating transformer-based sequence models, semi-supervised learning, domain adaptation, and uncertainty estimation may further improve robustness and practical reliability. Embedding the system into radiologist-in-the-loop workflows will be essential for real-world clinical translation.

Overall, the study demonstrates the significant potential of hybrid CNN–LSTM architectures as next-generation intelligent tools for assisting radiologists in precise and timely brain tumor diagnosis.

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